

Capture Analyzer操作说明

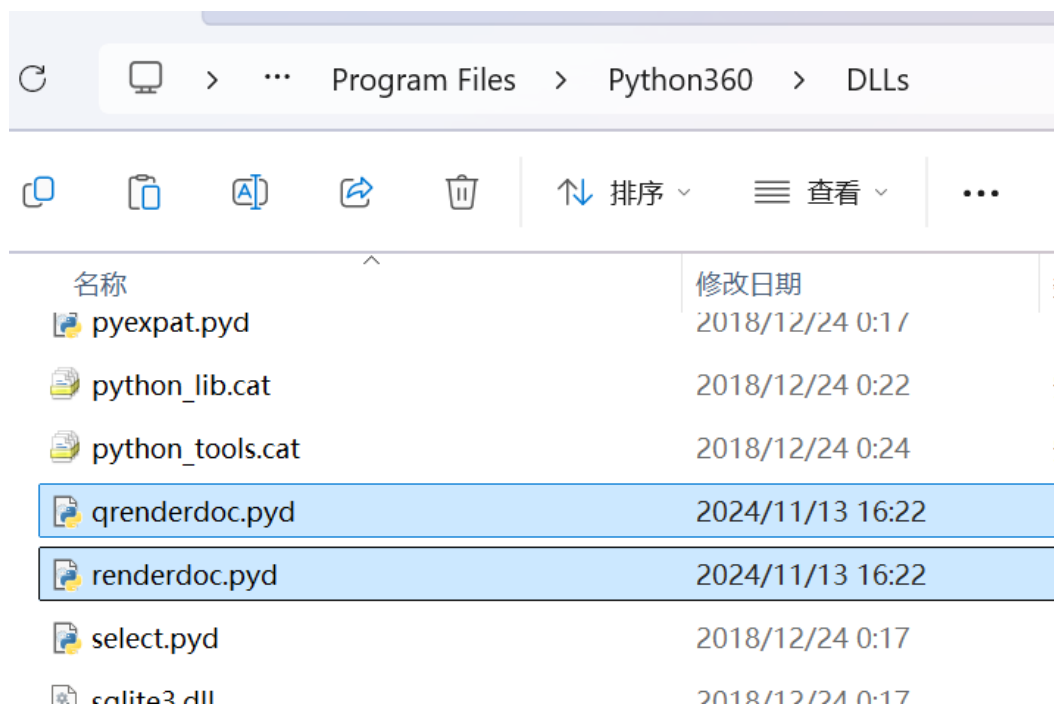
- 推荐创建一个空Unity工程执行还原操作，C#部分需要在Unity里执行，图方便可以把提供的整个文件夹放Unity工程内

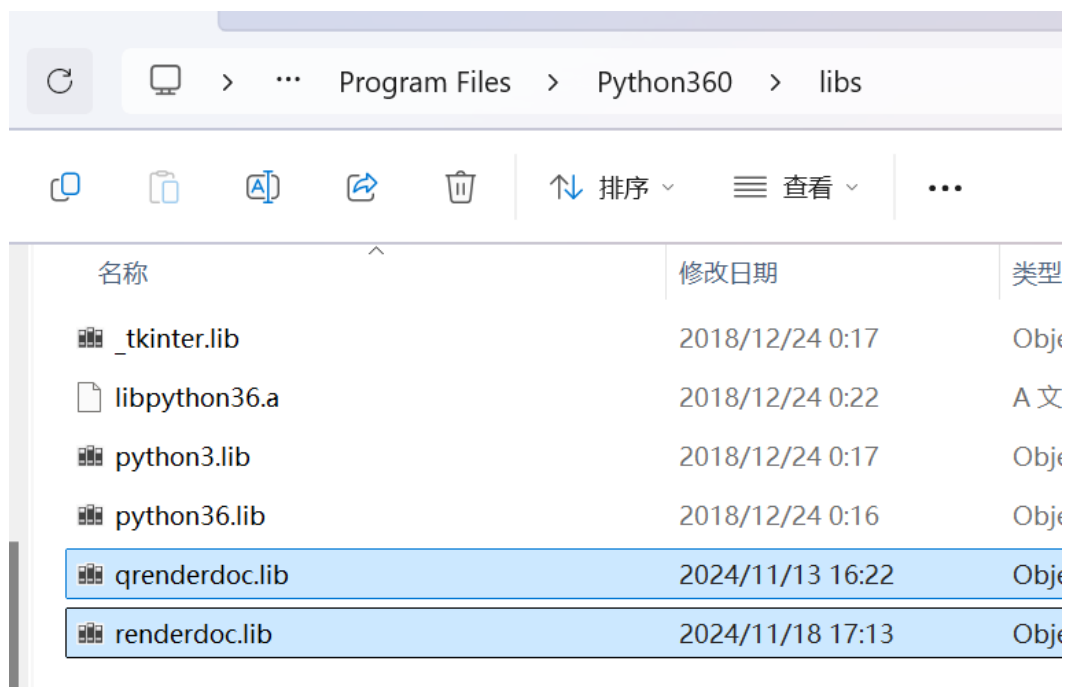
▼ 准备运行环境

▼ 在Renderdoc内进行

▼ 在Renderdoc外进行

1. 安装Pycharm
2. 安装Python **3.6.x (64bit)**
 - 安装时**必须勾选添加到环境变量**，或者**保证环境变量中PATH读取顺序里，3.6.x的路径在其他版本之前**
3. 将附带的lib放到python路径指定位置





▼ 导出数据

1. 打开DrawcallExporter.py设置如下部分

```
#####  
# Config Part #####  
#####  
filename = "E:/Inting/FrameCapture/AFK/MuMuVMMHeadLess_2024.11.13.15.35_frame2730.rdc"  
target_folder = "C:/Users/frankmli/Documents/Project/Moonflow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderCompiler/RenderDocDataExporter/Release"  
spirv_cross_path = "C:/Program Files/RenderDoc/plugins/spirv/spirv-cross.exe"  
  
# Because of read_mesh_data cost too much time, if you don't need to refresh extracted mesh data, set it to False to skip the process  
read_mesh_data = False  
|  
startEventID = 806  
endEventID = 1727  
  
unity_perDraw_Length = 20  
unity_instancing_perDraw_Length = 128
```

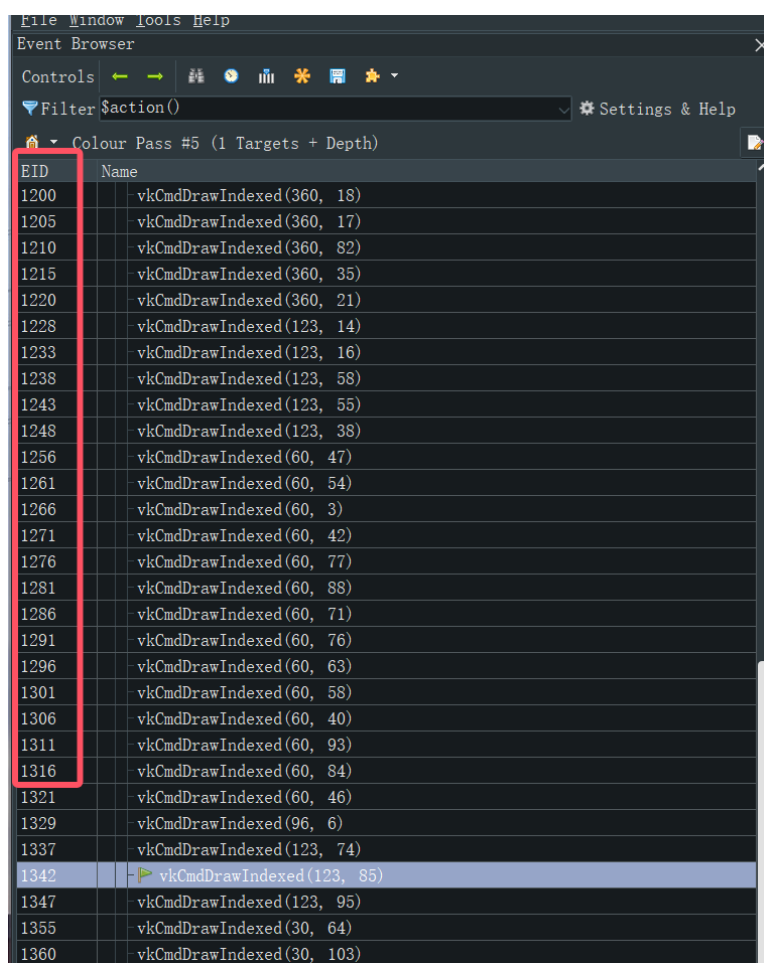
- filename：需要解包的.rdc截帧数据路径
- target_folder：将解包到该路径，为方便后续拼接操作，该路径可以考虑放置在用于还原的Unity工程的Assets路径内

▼ （可选）spirv_cross_path

- 用于翻译的spirv-cross.exe的位置，该插件附带在Renderdoc中。如配置路径则可以为所有Shader导出一个HLSL的编译版本（要求源码支持使用SPIRV-Cross）
- 该版本可以用于读取确认是否存在discard以在后续流程中设置生成材质的AlphaClip开关
- read_mesh_data: 设置为True时会导出模型数据为表格。由于导出模型数据耗时较高（目前导出速度大约在100个顶点数据/秒）且没有其他解决方

案，如果不需要更新导出模型资源时可以设置跳过

- startEventID\endEventID：设置数据的导出区间，数据来源于如下位置



EID	Name
1200	vkCmdDrawIndexed(360, 18)
1205	vkCmdDrawIndexed(360, 17)
1210	vkCmdDrawIndexed(360, 82)
1215	vkCmdDrawIndexed(360, 35)
1220	vkCmdDrawIndexed(360, 21)
1228	vkCmdDrawIndexed(123, 14)
1233	vkCmdDrawIndexed(123, 16)
1238	vkCmdDrawIndexed(123, 58)
1243	vkCmdDrawIndexed(123, 55)
1248	vkCmdDrawIndexed(123, 38)
1256	vkCmdDrawIndexed(60, 47)
1261	vkCmdDrawIndexed(60, 54)
1266	vkCmdDrawIndexed(60, 3)
1271	vkCmdDrawIndexed(60, 42)
1276	vkCmdDrawIndexed(60, 77)
1281	vkCmdDrawIndexed(60, 88)
1286	vkCmdDrawIndexed(60, 71)
1291	vkCmdDrawIndexed(60, 76)
1296	vkCmdDrawIndexed(60, 63)
1301	vkCmdDrawIndexed(60, 58)
1306	vkCmdDrawIndexed(60, 40)
1311	vkCmdDrawIndexed(60, 93)
1316	vkCmdDrawIndexed(60, 84)
1321	vkCmdDrawIndexed(60, 46)
1329	vkCmdDrawIndexed(96, 6)
1337	vkCmdDrawIndexed(123, 74)
1342	vkCmdDrawIndexed(123, 85)
1347	vkCmdDrawIndexed(123, 95)
1355	vkCmdDrawIndexed(30, 64)
1360	vkCmdDrawIndexed(30, 103)

▼（可选）unity_perDraw_Length

填入可能的UnityPerDraw 的长度标记。当CBuffer长度符合该设置时将被标识为UnityPerDraw。后续Unity内操作将在该Buffer中获取非Instancing渲染对象的unity_objectToWorld矩阵

▼（可选）unity_instancing_perDraw_Length

填入可能的UnityInstancing的长度标记。当CBuffer长度符合该设置时将被标识为UnityInstancing。后续Unity内操作将在该Buffer中获取Instancing渲染对象的unity_objectToWorld矩阵

2. 运行python等待导出。如能正常运行，应当会看到每个eventID会对应部分Log输出类似于

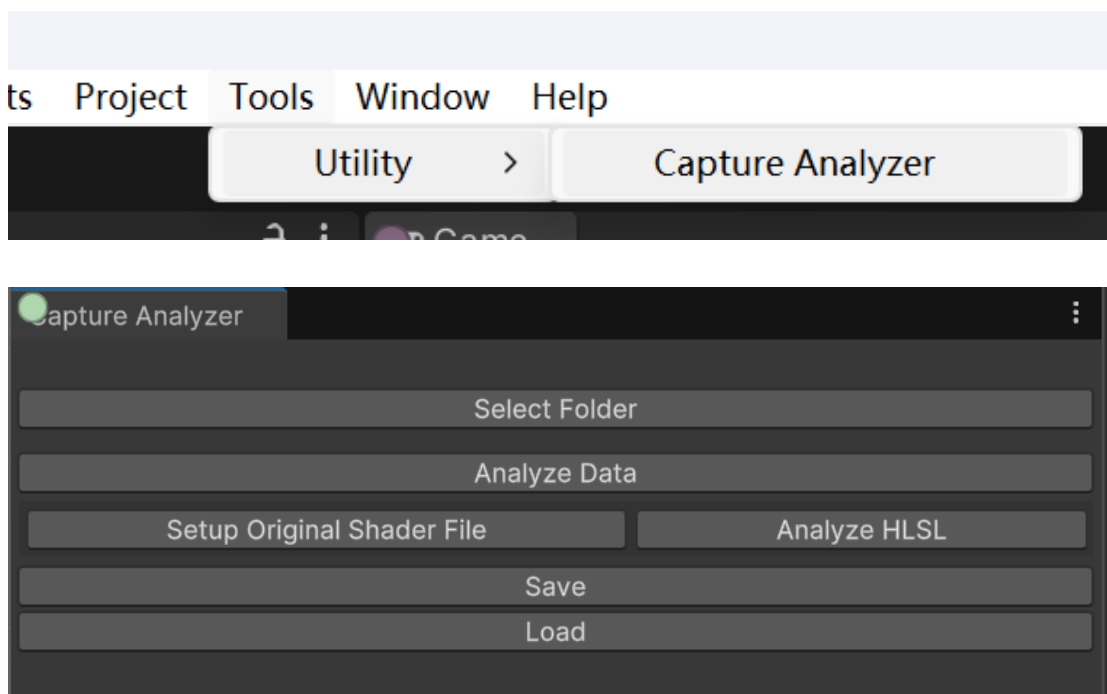
```

Action #128 Pass #0 starts with 811: vkCmdDrawIndexed()
Decoding Drawcall Data
Allowed disassembly targets:
SPIR-V (RenderDoc)
KHR_pipeline_executable_properties
AMD GCN ISA
disassemble_drawcall_setting
disassemble_vertex_shader
ShaderStage.Vertex CBuffers length: 3
Load Stage: Vertex CBuffer: 0 Name: ResourceId::12951 Shader:ResourceId::53281
Load Stage: Vertex CBuffer: 1 Name: ResourceId::17167 Shader:ResourceId::53281
Load Stage: Vertex CBuffer: 2 Name: ResourceId::17133 Shader:ResourceId::53281
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Vertex_s2_b0_res17_16670
disassemble_pixel_shader
ShaderStage.Pixel CBuffers length: 4
Load Stage: Pixel CBuffer: 0 Name: ResourceId::59666 Shader:ResourceId::53282
Load Stage: Pixel CBuffer: 1 Name: ResourceId::12951 Shader:ResourceId::53282
Load Stage: Pixel CBuffer: 2 Name: ResourceId::17167 Shader:ResourceId::53282
Load Stage: Pixel CBuffer: 3 Name: ResourceId::17133 Shader:ResourceId::53282
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b1_res18_52396
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b2_res19_52285
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b3_res20_50847
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b4_res21_7583
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b5_res25_9211
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b6_res26_16720
Save Texture: C:/Users/frankmhl/Documents/Project/MoonFlow/TempMF/Assets/moonflow-system/Tools/Editor/MFUtilityTools/MFShaderDecompiler/RenderDocDataExporter/Release/811/Pixel_s2_b7_res27_16691
Combining through 12 indices
Index 0 / 12
Index 1 / 12
Index 2 / 12
Index 3 / 12
Index 4 / 12
Index 5 / 12
Index 6 / 12
Index 7 / 12
Index 8 / 12
Index 9 / 12
Index 10 / 12
Index 11 / 12
Writing 12 vertices
Writing 12 indices
Writing to disk
当前时间: 2024-12-11 11:00:39

```

▼ 拼接数据

1. 打开Unity，通过Tool - Utility - CaptureAnalyzer打开拼接工具



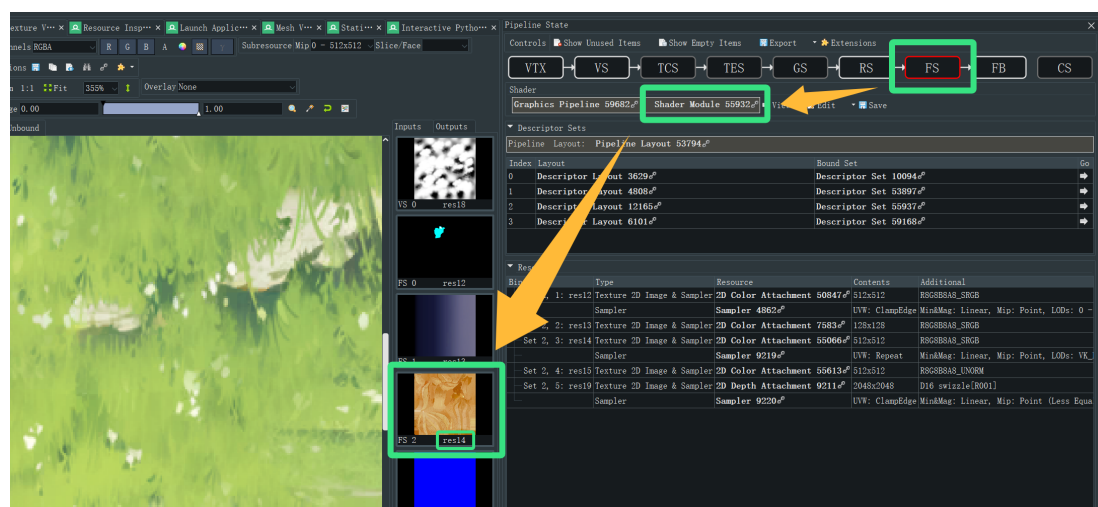
2. 点击Select Folder选择目标路径，即上文中的target_folder
3. 点击Analyze Data进行数据分析，这个操作可能会需要几秒钟到十几秒钟，中间Unity会卡住

- 上一步完成后点击Setup Original Shader File，这一步将会重新组合Vertex Shader与Pixel Shader，最终得到该段截帧数据实际用到的shader列表，并显示在界面中

Select Folder			
Analyze Data			
Setup Original Shader File		Analyze HLSL	
Vertex Shader:	53307	Pixel Shader:	53308 0
Vertex Shader:	51267	Pixel Shader:	51268 0
Vertex Shader:	55931	Pixel Shader:	55932 0
Vertex Shader:	53795	Pixel Shader:	53796 0
Vertex Shader:	51277	Pixel Shader:	51278 0
Vertex Shader:	53802	Pixel Shader:	53803 0
Vertex Shader:	52150	Pixel Shader:	52151 0
Vertex Shader:	51365	Pixel Shader:	51366 0
Vertex Shader:	53600	Pixel Shader:	53601 0
Vertex Shader:	51289	Pixel Shader:	51290 0
Vertex Shader:	51297	Pixel Shader:	51298 0
Vertex Shader:	51155	Pixel Shader:	51156 0
Vertex Shader:	55453	Pixel Shader:	55454 0
Vertex Shader:	53332	Pixel Shader:	53333 0
Vertex Shader:	54915	Pixel Shader:	54916 0
Vertex Shader:	55967	Pixel Shader:	55968 0
Vertex Shader:	51166	Pixel Shader:	51167 0
Vertex Shader:	53343	Pixel Shader:	53344 0
Vertex Shader:	53921	Pixel Shader:	53922 0
Vertex Shader:	53349	Pixel Shader:	53350 0
Vertex Shader:	53281	Pixel Shader:	53282 0
Vertex Shader:	53565	Pixel Shader:	53566 0
Vertex Shader:	53298	Pixel Shader:	53299 0
Save			
Load			

- (可选) 该Shader列表右侧结尾有一个可填序号，默认是0，其值为自动绑定diffuse贴图资源的序号。该值为0时，将不会自动贴diffuse贴图。该值为-1时，将不会在后续Load步骤中加载模型时跳过该shader（一般用于跳过描边、阴影等shader pass）。需要自动赋值Diffuse贴图时请在如下位置人工确认diffuse来源

下图样例中意为Pixel Shader的ID为55932的Shader（上图列表从上往下第三个）的Diffuse贴图的ResourceID为14，则列表内VertexShader 55931\PixelShader 55932的项末尾填14



6. （可选）如果前文步骤中导出了HLSL版本的shader，此时可以点击Analyze HLSL File重新拼接一个符合Unity读取的新版本HLSL shader，不过目前暂时无法使用
7. 点击Save将重组数据保存为永久资源，每个Event文件夹内会新建一个Translated文件夹并将该Event的重组资源存入该文件夹
8. 点击Load依照前文规则在**当前场景**中重组模型实例。